

INDEPENDENT PEER-REVIEW PACKET

P02-LM-002

**Exact Determinant-Weight r^4 Second-Moment
Interface**

Cold Review Packet 17 of 20

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Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	17 of 20
Object ID	P02-LM-002
Object class	Algebraic/probabilistic interface
Exact statement	For the exact endpoint Hessians and typed weight W_r , $W_{r^2} \leq 16 r^4 R^8$ for $0 < r \leq 1$; if $\sup E[R^8] = M_8 < \infty$ then $E[W_{r^2}] \leq 16 M_8 r^4$.
Internal terminal label	CLOSED - EXACT UNIFORM r^4 DETERMINANT-WEIGHT SECOND-MOMENT INTERFACE
Terminal date	2026-07-24
Recommended expertise	Matrix algebra / probability inequalities / Kac-Rice weights

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Exact Hessian factorization, pointwise domination, and independent symbolic/computational audit.
Prior review record	Anthropic independently checked the factorization, r power, and R^8 bound.
Binding exclusions	Does not prove the eighth-moment premise for any law, normalizer bounds, Palm transfer, or P0.2.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: Does not prove the eighth-moment premise for any law, normalizer bounds, Palm transfer, or P0.2.

4. Object-specific hostile checks

- Reconstruct determinant factorizations from the endpoint Hessian convention.
- Check the indicator functions only reduce the weight.
- Verify the expectation corollary keeps the uniform eighth-moment premise explicit.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

LCR-CAP-011-v1.0 — P02-LM-002 CLOSED EXACT DETERMINANT-WEIGHT r^4 MOMENT INTERFACE

PEER-REVIEW CAPSULE AND TERMINAL RECORD

Artifact ID: LCR-CAP-011-v1.0
Lemma ID: P02-LM-002
Closure ID: LCR-CLS-010-v1.0
Prepared and adjudicated by: ChatGPT — OpenAI GPT-5.6 Thinking, Lemma Closure Director
Date: 2026-07-24
Class: CAP / CLS / EXACT ALGEBRAIC-MOMENT INTERFACE
Authority: terminal lemma record within exact scope under OP-PROT-011
Canonical mathematical impact: NONE
Terminal label: CLOSED — EXACT UNIFORM r^4 DETERMINANT-WEIGHT SECOND-MOMENT INTERFACE

[[OBJ:P02-LM-002]] [[STATUS:CLOSED-EXACT]] [[DM2:CLOSED]] [[LIMITING-CONSTANT:EXCLUDED]] [[P0.2:OPEN]]

1. READ FIRST

One-sentence result:

For the exact normalized endpoint-Hessian convention, the squared typed determinant weight satisfies the pointwise inequality $W_r^2 \leq 16r^4 R^8$ for every $0 < r \leq 1$; consequently any family of jet laws with uniformly bounded eighth R-moment satisfies $E[W_r^2] \leq 16M_8 r^4$.

Why it matters:

This is the precise DM2 interface required to transfer super-polynomial conditioned-Gaussian tail bounds through determinant-weighted Palm reweighting. It isolates the exact r^4 algebra from the separate Gaussian regularity, normalizer, and GT5 lemmas.

Primary proof:

GP-DER-071-v1.0, Sections 1–2.
Drive ID: 1MPY74ihf6hxrr97d7nnK-9Id1ARI3xL_2Eb63HdpV8Y.

Independent audit:

CL-AUD-052-v1.0.
Drive ID: 1kqZ2KB5ysCHZiVueNaVfIKJfAsljF0W3HagoMVNT5Yw.

Object crosswalk:

CL-AUD-045-v1.0, Drive ID 1mhNO8GXWqxeMnqFyE5xpyZvr_OdvGomJ_EsDeUjnUxY.
CL-AUD-046-v1.0, Drive ID 1NvD8kYTNBX2ZP27s8mr6NfnwbIWk9-kwgBvwjxLpBVw.

Current status:

CLOSED — EXACT UNIFORM r^4 DETERMINANT-WEIGHT SECOND-MOMENT INTERFACE.

P0.2 remains open.

2. FROZEN EXACT STATEMENT

Let $0 < r \leq 1$ and let

$(q, a, w, z, c_{40}, c_{31}, c_{22}, c_{13}, c_{04})$ in $\mathbb{R}^9$.

Define

$$R = 1 + |q| + |a| + |w| + |z| + |c_{40}| + |c_{31}| + |c_{22}| + |c_{13}| + |c_{04}|.$$

Define the exact endpoint Hessians

$H_M =$

$$\begin{bmatrix} -r + r^2 c_{40}/12, & -ar/2 + r^2 c_{31}/12, \\ -ar/2 + r^2 c_{31}/12, & q - rw/2 + r^2 c_{22}/8 \end{bmatrix},$$

$H_S =$

$$\begin{bmatrix} r + r^2 c_{40}/12, & ar/2 + r^2 c_{31}/12, \\ ar/2 + r^2 c_{31}/12, & q + rw/2 + r^2 c_{22}/8 \end{bmatrix}.$$

Let

$$I_M = 1\{H_M \text{ is negative definite}\}, \\ I_S = 1\{\det H_S < 0\},$$

and define the nonnegative typed determinant weight

$$W_r = |\det H_M \det H_S| I_M I_S.$$

THEOREM P02-LM-002.

For every such jet vector and every $0 < r \leq 1$,

$$W_r^2 \leq 16r^4 R^8.$$

Consequently, for any probability law μ_r on the jet vector for which

$$M_8 := \sup_{0 < r \leq r_0} E_{\mu_r}[R^8] < \infty$$

for some $0 < r_0 \leq 1$, one has uniformly for $0 < r \leq r_0$,

$$E_{\mu_r}[W_r^2] \leq 16M_8 r^4.$$

No Gaussian assumption is needed for the pointwise inequality. The probability-law corollary uses only the displayed eighth-moment premise.

3. EXACT PROOF

Define

$$\begin{aligned} A_M &= -1 + rc40/12, \\ B_M &= -a/2 + rc31/12, \\ C_M &= q - rw/2 + r^2c22/8, \end{aligned}$$

$$\begin{aligned} A_S &= 1 + rc40/12, \\ B_S &= a/2 + rc31/12, \\ C_S &= q + rw/2 + r^2c22/8. \end{aligned}$$

Then exactly

$$\begin{aligned} H_M &= [[rA_M, rB_M], [rB_M, C_M]], \\ H_S &= [[rA_S, rB_S], [rB_S, C_S]]. \end{aligned}$$

Therefore

$$\begin{aligned} \det H_M &= rD_M, \\ \det H_S &= rD_S, \end{aligned}$$

where

$$\begin{aligned} D_M &= A_M C_M - rB_M^2, \\ D_S &= A_S C_S - rB_S^2. \end{aligned}$$

Since the indicators are idempotent and bounded by one,

$$\begin{aligned} W_{r^2} &= r^4 D_M^2 D_S^2 I_{MI_S} \\ &\leq r^4 D_M^2 D_S^2. \end{aligned}$$

For $0 < r \leq 1$, every coefficient appearing in $A_M, A_S, B_M, B_S, C_M, C_S$ has absolute value at most one and every jet absolute value is included in R . Hence

$$|A_M|, |A_S|, |B_M|, |B_S|, |C_M|, |C_S| \leq R.$$

Thus

$$\begin{aligned} |D_M| &\leq |A_M C_M| + r |B_M|^2 \leq 2R^2, \\ |D_S| &\leq |A_S C_S| + r |B_S|^2 \leq 2R^2. \end{aligned}$$

Therefore

$$W_{r^2} \leq r^4 (2R^2)^2 (2R^2)^2 = 16r^4 R^8.$$

Taking expectations gives the stated corollary.

4. PROOF-OBLIGATION LEDGER

M1 — Endpoint-Hessian factorization.

Status: PASS EXACTLY.

M2 — Exact r^4 factor after squaring.

Status: PASS EXACTLY. No regression or asymptotic fitting is used.

M3 — Indicator handling.

Status: PASS. $I_M^2 = I_M$ and $I_S^2 = I_S$; both are at most one.

M4 — Uniform coefficient domination for $0 < r \leq 1$.

Status: PASS.

M5 — D_M, D_S quadratic bound.

Status: PASS.

M6 — Pointwise W_r^2 bound.

Status: PASS.

M7 — Moment corollary.

Status: PASS under exactly the stated M_8 premise.

M8 — Measure independence.

Status: PASS. No density, Gaussianity, or Palm normalization is used in the implication from M_8 to the second-moment bound.

5. INDEPENDENT REVIEW

CL-AUD-052 independently checked:

- - the endpoint-Hessian factorization in exact rational arithmetic;
- - the determinant factor r at each endpoint;
- - the exact r^4 factor in W_r^2 ;
- - the absence of any lower r -power;
- - the pointwise R^8 domination by hand and over 200,000 random draws;
- - the consistency of the endpoint object with the declared fold frame and downstream structures.

Review family: Anthropic/Claude, organizationally distinct from the OpenAI proof author.

Review classification:

ORGANIZATIONALLY DISTINCT / SYMBOLIC-COMPUTATIONAL AUDIT / ORIGINAL PROOF VISIBLE.

The review is not clean-room in the strict sense, but it independently recomputed every claim included in this frozen lemma. It counts as one independent-eyes credit.

CL-AUD-045 and CL-AUD-046 separately crosswalked the normalized Hessians, jets, six-pin construction, pair-Palm object, and branch/type conventions. Their remaining historical open edges concern broader P0.2 integration, not the pointwise algebra frozen here.

6. EDGE CASES

$r=1$:

The coefficient domination remains valid because all r and r^2 coefficients are at most one.

$r>0$:

The inequality remains valid for every positive r . No division by r occurs in the proof.

Zero jets:

$R=1$ and the formula is well-defined.

Type-boundary cases:

If either indicator vanishes, $W_r=0$ and the bound is immediate. No assumption that the type boundary has zero probability is required for this lemma.

Singular Hessians:

Permitted. They only reduce the typed weight.

Arbitrary law:

Permitted for the pointwise result. The expectation conclusion requires only the finite uniform eighth moment stated explicitly.

7. EXCLUSIONS

This closed lemma does not establish:

- - that a specific exact-torus Gaussian jet family has uniform eighth moments;
- - the limiting value of $E[W_r^2]/r^4$;

- $E[Q_L^4 \mathbf{1}_{\{Q_L < 0\}}]$;

- - the planar truncated-normal closed form or decimal;
- - a normalizer lower or upper bound;
- - a Palm tail estimate without the separate normalizer premise;
- - the repaired GT5 theorem;
- - exact-field capture;
- - $1-a_r=O(r^3)$;
- - P0.2 terminal status.

Those statements may cite this lemma but require additional dependencies.

8. CRITIQUE HISTORY

Earlier CL-AUD-047 treated DM2 partly computationally. GP-DER-071 supplied the exact symbolic proof. CL-AUD-052 independently verified the exact factorization and domination and upgraded the component from Monte Carlo corroboration to symbolic proof plus audit.

No mathematical correction to the frozen P02-LM-002 statement was required during this closure pass.

The broader GP-DER-071 document also discusses the exact-torus limiting constant and planar formula. Those are deliberately excluded because their proof and review obligations differ from the minimal DM2 interface.

9. CLOSURE DECISION

Closure ID: LCR-CLS-010-v1.0

Lemma ID: P02-LM-002

Date: 2026-07-24

Terminal label:

CLOSED — EXACT UNIFORM r^4 DETERMINANT-WEIGHT SECOND-MOMENT INTERFACE.

Decisive evidence:

- - one-line exact determinant factorization;
- - exact pointwise polynomial domination;
- - organizationally distinct symbolic/computational audit;
- - object crosswalk to the declared exact endpoint convention;
- - complete scope and edge-case fence.

OP-PROT-011 independent-eyes condition is satisfied. This is an exact interface lemma, not canonical promotion of P0.2.

10. DOWNSTREAM USAGE

Authorized use:

If a downstream exact jet law proves $\sup E[R^8] \leq M_8$, it may cite P02-LM-002 to conclude

$$E[W_r^2]^{(1/2)} \leq 4\sqrt{M_8} r^2.$$

Combined with a separate normalizer lower bound $Z_r \geq c_Z r^2$, Cauchy-Schwarz gives

$$\frac{E[W_{r1_E}]}{Z_r} \leq 4\sqrt{M_8} P(E)^{(1/2)} / c_Z.$$

This displays precisely how a conditioned-law tail transfers to the typed determinant-weighted law.

The lemma supplies no bound on $P(E)$ and no normalizer bound.

11. REOPENING CONDITIONS

Reopen if:

- - an endpoint Hessian coefficient or jet convention is corrected;
- - the determinant factorization is shown wrong;
- - r is used outside $0 < r \leq 1$ without revising the coefficient bound;
- - R is redefined without including every displayed jet;
- - a downstream source cites the expectation result without the eighth-moment premise.

Do not reopen because a sharper constant, better moment norm, or limiting constant is found.

12. MANIFEST

Primary proof:

- - GP-DER-071-v1.0
- - Drive ID 1MPY74ihf6hxr97d7nnK-9Id1ARI3xL_2Eb63HdpV8Y

Independent audit:

- - CL-AUD-052-v1.0
- - Drive ID 1kqZ2KB5ysCHZiVueNaVfIKJfAsljF0W3HagoMVNT5Yw

Object crosswalk:

- - CL-AUD-045-v1.0
- - Drive ID 1mhNO8GXWqxeMnqFyE5xpyZvr_OdvGomJ_EsDeUjnUxY
- - CL-AUD-046-v1.0
- - Drive ID 1NvD8kYTNBX2ZP27s8mr6NfnwbIWk9-kwgBvwjxLpBVw

Closure capsule:

- - LCR-CAP-011-v1.0 / LCR-CLS-010-v1.0
- - Drive ID 1P84b7pNI-z8l9jQrRWW9pP5dNcA4TszKlhMZDFxBIWw

13. FINAL CHECKLIST

Exact statement: PASS.

Proof complete: PASS.

Quantifiers and r -domain explicit: PASS.

Moment premise explicit: PASS.

Object convention crosswalked: PASS.

Independent review: PASS.

Edge cases: PASS.

No circular dependency: PASS.

Scope fence: PASS.

P0.2 separation: PASS.

FINAL VERDICT:

APPROVE AND CLOSE P02-LM-002 WITHIN THE EXACT POINTWISE AND CONDITIONAL MOMENT SCOPE.

END LCR-CAP-011-v1.0

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT UNIFORM r^4 DETERMINANT-WEIGHT SECOND-MOMENT INTERFACE
Closure date	2026-07-24
Evidence basis	Exact Hessian factorization, pointwise domination, and independent symbolic/computational audit.
Independent review already recorded	Anthropic independently checked the factorization, r power, and R^8 bound.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- Does not prove the eighth-moment premise for any law, normalizer bounds, Palm transfer, or P0.2.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: <https://docs.google.com/document/d/1P84b7pNI-z8l9jQrRWW9pP5dNcA4TszKIhMZDFxBIWw/edit>

Closure / terminal record: <https://docs.google.com/document/d/1P84b7pNI-z8l9jQrRWW9pP5dNcA4TszKIhMZDFxBIWw/edit>

Local source snapshot used to construct this packet: LCR-CAP-011-v1.0 — P02-LM-002 Closed Exact Determinant-Weight r^4 Moment Interface

Local source snapshot SHA-256 (DOCX export, transport checksum only):
4bd80ff069ca37de5e003c50ba7cb0e6efbd3d3cce0661bba46cda7ceb2a1710

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.